Alexa Kamm

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EXPERIENCE

Software Developer, ***Freddie Mac,*** McLean, VA **06/2024 – Present**

- Design and develop a batch-processing pipeline to load transactional data from Aurora PostgreSQL into S3-backed Apache Iceberg tables, optimizing SQL queries, memory utilization, and data visibility to support reliable, high-performance processing.
- Serve as on-call engineer for an enterprise data platform supporting hundreds of users; perform root cause analysis using Splunk, Elasticsearch, and Kibana, and implement preventive fixes, resolving 250+ incidents/tickets while maintaining a same-day SLA for production incidents and a 2-day SLA for non-production.
- Re-architected Spark-based data ingestion pipeline parameter management by migrating from Delta Lake (S3) to AWS DynamoDB, reducing configuration retrieval latency by ~60–80%; provided a reusable Python library for these DynamoDB CRUD operations, standardizing configuration management code.
- Led migration of Snowflake authentication from CyberArk password-based access to OAuth 2.0, eliminating long-lived secrets and aligning with enterprise security and IAM best practices.
- Developed RESTful APIs and AWS Lambda functions for job orchestration, status tracking, and data movement within distributed pipeline architecture.
- Designed backward and forward compatibility across 6 versions of an internal code library, ensuring API stability, enabling zero-downtime deployments, and preventing breaking changes across 10 lines of business; implemented dynamic dependency management for third-party libraries, reducing failures during platform upgrades and improving system reliability.
- Built an AWS Glue Data Catalog inter-region synchronization system to support disaster recovery, ensuring high availability of over 800 production databases' table and partition metadata across regions.

Special Events Coordinator (Volunteer), ***Remote Area Medical at UVA,*** Charlottesville, VA **02/2022 – 05/2024**

- Secured $30,000 in grant funding for a student-run free healthcare clinic serving underserved populations.
- Organized large-scale 5K fundraiser by managing permits, liability insurance, and institutional funding.

Maternal Fetal Medicine Clinical Research Intern, ***UVA Health Systems,*** Charlottesville, VA **05/2022 – 05/2023**

- Led patient consent discussions and independently screened/enrolled participants in clinical trials.
- Collaborated with healthcare providers to evaluate patient eligibility and ensure protocol compliance.
- Managed study databases, processed biological samples, and reported progress to study sponsors.

Front-End Developer, ***The Virginia Discourse Lab,*** Charlottesville, VA **10/2022 – 12/2022**

- Built a user interface using HTML/CSS within a Django framework to support a searchable database of literary dialogue (Jane Austen corpus).

Technical Skills

- **Languages:** Python, Scala, Java, Bash, SQL
- **Data Engineering:** Apache Spark, Hadoop, Delta Lake, Apache Iceberg
- **Cloud & DevOps:** AWS (Lambda, DynamoDB, S3, EMR, EMR on EKS, EC2, EKS, Step Functions, CloudFormation, API Gateway, VPC, Glue), Kubernetes, Karpenter, kubectl, Docker, Jenkins (CI/CD)
- **Databases & Storage:** Snowflake, Aurora Postgres, MongoDB
- **Data Formats:** Parquet, JSON, YAML, XML, CSV
- **Tools:** Git, Bitbucket, Jira, Splunk, Elasticsearch, Kibana, Jupyter Notebook
- **AI-Assisted Development:** Kiro (Spec-driven development, workflows, steering, NLP)

EDUCATION, AWARDS & CERTIFICATIONS

University of Virginia, Charlottesville, VA **05/2024**

- B.A., Computer Science & English Literature | GPA: 3.95
- **Honors:** Phi Beta Kappa; Michael Wagenheim Memorial Scholarship; William and Charlotte Savage Scholarship

AWS Certified Cloud Practitioner (CLF-C02) **06/2025**
